Analyst

PAPER

Cite this: *Analyst*, 2014, **139**, 3179

Acquisition of reproducible transmission near-infrared (NIR) spectra of solid samples with inconsistent shapes by irradiation with isotropically diffused radiation using polytetrafluoroethylene (PTFE) beads

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A bead-incorporated transmission scheme (BITS) has been demonstrated for collecting reproducible transmission near-infrared (NIR) spectra of samples with inconsistent shapes. Isotropically diffused NIR radiation was applied around a sample and the surrounding radiation was allowed to interact homogeneously with the sample for transmission measurement. Samples were packed in 1.40 mm polytetrafluoroethylene (PTFE) beads, ideal diffusers without NIR absorption, and then transmission spectra were collected by illuminating the sample-containing beads using NIR radiation. When collimated radiation was directly applied, a small portion of the non-fully diffused radiation (NFDR) propagated through the void space of the packing and eventually degraded the reproducibility. Pre-diffused radiation was introduced by placing an additional PTFE disk in front of the packing to diminish NFDR, which produced more reproducible spectral features. The proposed scheme was evaluated by analyzing two different solid samples: density determination for individual polyethylene (PE) pellets and identification of mining locality for tourmalines. Because spectral collection was reproducible, the use of the spectrum acquired from one PE pellet was sufficient to accurately determine the density of nine other pellets with different shapes. The differentiation of tourmalines, which are even more dissimilar in appearance, according to their mining locality was also feasible with the help of the scheme.

Received 9th October 2013
Accepted 26th March 2014

DOI: 10.1039/c3an01904g

www.rsc.org/analyst

1. Introduction

Diffuse reflectance near-infrared (NIR) spectroscopy has been extensively utilized for non-destructive analysis of diverse solid samples, such as grains[1–3] and polymers,[4,5] because it does not require further extensive sampling steps. Samples are usually packed in a container and rotated during spectral collection to improve sample representation. Analysis of individual solid samples is also in great demand, such as for the quality assurance of individual pharmaceutical tablets.[6–8] When a single solid sample is subjected to NIR spectroscopic analysis, both diffuse reflectance and transmission spectral collection schemes are available. Diffuse reflectance measurements result in spectra containing more weighted chemical information on constituents localized near sample surfaces; therefore, acquired spectral information occasionally fails to fully represent the whole sample composition, especially when a sample is internally inhomogeneous. NIR transmission measurement is preferable for correct compositional sample representation and accurate analysis, as demonstrated in several previous publications.[9–15]

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When a sample with a consistent shape, such as pharmaceutical tablets, is analyzed using NIR spectroscopy, reproducible transmission spectra are easily collected using a simple sample holder. Conversely, the transmission spectral features of samples with dissimilar shapes vary substantially in the case of irradiating collimated radiation, resulting in spectra with poor reproducibility. Therefore, a spectral collection strategy that can produce reproducible transmission spectra for these samples is necessary. One possible approach is application of diffused radiation around a solid sample for transmission spectral collection, rather than using collimated or focused NIR radiation. When diffused radiation is applied around the sample, different types of photon migrations occur across the sample. Under this circumstance, the overall interaction of diffused radiation with the sample does not vary significantly even though its shape varies, thereby enabling the collection of reproducible transmission spectra.

In this publication, we demonstrate a bead-incorporated transmission scheme (BITS) in which 1.40 mm polytetrafluoroethylene (PTFE) beads are used to surround a target sample to generate isotropically diffused NIR radiation for transmission measurements. PTFE beads diffuse incident NIR radiation in a nearly random fashion without self-absorption.[16,17] The diffused

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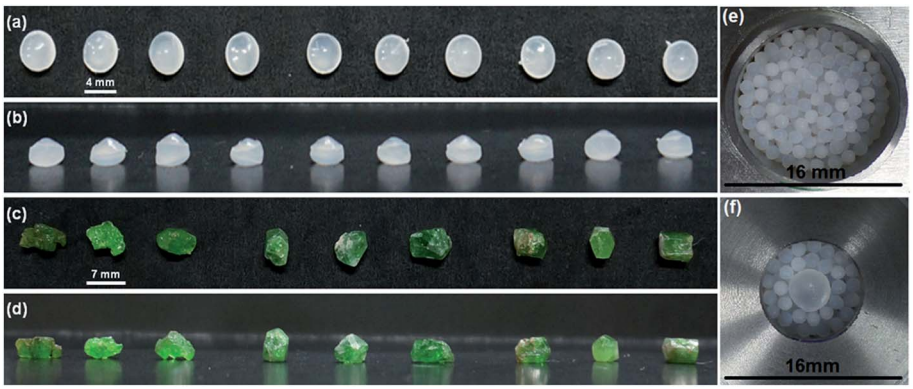

Fig. 1 Pictures of 10 PE pellets of the same grade ((a) top-view and (b) side-view) and tourmalines from three different mining localities ((c) top-view and (d) side-view). Front- (e) and back-views (f) of the 1.40 mm PTFE bead packing containing a PE pellet are also shown.

radiation fully interacts with the sample even if its shape varies, and the interacted radiation is gathered using a lens to produce transmission spectra. After each spectral acquisition, the beads are separated easily by sieving. The BITS has been used for the analysis of two different solid samples: density determination of individual polyethylene (PE) pellets and identification of tourmaline mining locality. Tourmalines are semi-precious stones of mainly boron silicate compounded with Al, Fe, Mg, Li and K. Fig. 1 shows real pictures of 10 PE pellets of the same grade and some of the tourmalines used in the study. In both cases, sample shapes were different from each other, and tourmalines were more dissimilar in appearance.

The first study initially evaluated the reproducibility of replicate transmission spectral features collected for a PE pellet surrounded by PTFE particles with four different sizes (0.20, 0.35, 0.76 and 1.40 mm). Then, under the determined optimal experimental conditions, NIR transmission spectra of 10 pellets randomly selected from each PE sample were individually collected. Partial least squares (PLS)[18,19] was used to model density variation by incorporating only one pellet spectrum out of ten pellet spectra. Densities of the pellets from validation samples were predicted using the PLS model, and the resulting accuracies were evaluated. Variations in accuracy were investigated when the number of pellets from each calibration PE sample included for PLS changed from one to nine. In the second study, NIR transmission spectra of tourmalines from three different mining localities were collected and used to identify their mining localities by principal component analysis (PCA)[19,20] combined with linear discriminant analysis (LDA).[21–23] Replicate spectra collected by randomly changing the orientation of tourmaline were also examined in parallel.

2. Experimental section

2.1 PE pellets and tourmalines

Twenty-five PE samples with different densities (0.918–0.961 g cm^{-3}) used in a previous publication[24] were also employed in this study. Fig. 1(a) and (b) show top- and side-view pictures of 10 randomly selected pellets from a PE sample (density: 0.918 g cm^{-3}), respectively. The overall shapes were disk-like and generally similar, while the detailed shapes were fairly different from each other. Out of the 25 PE samples, 20 samples were used as a calibration set and 5 independent samples were used as a validation set.

Tourmalines obtained from three different mining regions (referred to as tourmaline A, B and C; 5 samples from each locality) in Myanmar were used. A reliable gemstone researcher provided the samples for this study. Fig. 1(c) and (d) show top- and side-view pictures, respectively, of 9 tourmalines (3 samples from each locality). Tourmaline shapes varied more than those of PE pellets, and their mining localities could not be distinguished by visual observation.

2.2 Acquisition of transmission NIR spectra using BITS

A variable pathlength cell with a diameter of 16 mm was used for the BITS-based measurement. In this study, four different sizes of 0.20 mm, 0.35 mm, 0.76 mm, and 1.40 mm PTFE particles were used to construct a packing around a pellet. PTFE particles were partially packed initially and the pellet was positioned at the center. Extra particles were then added around the pellet and the whole packing was gently pressed with a glass window for spectral collection. Pictures showing front and back views of a PE pellet with 1.40 mm particle (bead) packing are shown in Fig. 1(e) and (f), respectively. To collect transmission spectra, NIR radiation was applied towards the front of a sample-containing bead packing (Fig. 1(e)). Sample-interacted radiation was then gathered using a lens and delivered to an InGaAs detector (FT-NIR spectrometer, ABB Bomem). The resolution of the collected spectra was 8 cm^{-1}.

All spectral pre-processing and multivariate analysis, including baseline correction, normalization, PLS, PCA and LDA, were performed using MATLAB Version 7.0 (Math Works Inc., MA, USA). Monte-Carlo simulation was performed using Montcarl Version 2013 (a free computational package).[25,26]

3. Results and discussion

3.1 Transmission NIR spectral features of a PE pellet in PTFE particles of different sizes

The characteristics of photon propagation in a packing vary depending on the particle size and the amount of particles. We

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have therefore investigated variation in the transmission spectral features of a PE pellet when it is placed in the packing of PTFE particles of four difference sizes of 0.20 mm, 0.35 mm, 0.76 mm, and 1.40 mm. In the cases of using 0.20 mm, 0.35 mm, and 0.76 mm particles, the total amount of the particles was determined to be 0.40 g, the minimum quantity to fully surround a pellet in a transmission cell. When the 1.40 mm particles were used, it was necessary to increase the amount to 0.72 g due to the larger size.

Fig. 2(a) shows five replicate transmission NIR spectra of a PE pellet (density: 0.918 g cm^{-3}) positioned in a packing composed of PTFE particles of four different sizes (0.20 mm, 0.35 mm, 0.76 mm, and 1.40 mm). Baseline-corrected spectra in the 8800–6300 cm^{-1} range, linearly offset at 8800 cm^{-1}, 7480 cm^{-1} and 6300 cm^{-1}, are shown in the figure. The wide face of the pellet was perpendicularly positioned with respect to the NIR illumination for spectral collection. Replicate spectra were collected by repeatedly reloading the pellet and filling the particles at each measurement. To collect the background single beam spectrum, the same amount of particles without a pellet was used. The most obvious observation is that the absorption magnitude of the pellet increases as the size of the surrounding particles in a packing becomes smaller. Since a packing composed of smaller particles is denser, the diffusion of radiation is increased. The more diffused photons have greater interaction with the pellets, so leading to greater absorption. Photon propagation crossing a PTFE-particle packing will never be consistent when the packing is randomly re-packed. Individually diffused photons travel different paths inside the pellet at each replicate measurement, and the magnitude of absorption can subsequently vary as observed. As shown in Fig. 2(a), variation in the absorbance among the replicate spectra is greater when the particles in a packing are smaller, such as the greatest variation in the measurement based on a 0.2 mm particle-packing. Increased diffusion is advantageous because it enhances the sensitivity of the absorption-based measurement; however, superfluous photon diffusion adversely impacts the reproducibility of the measurement due to the greater attenuation of the photons as well as the increased complexity in individual photon migration.

To compensate for the variation in absorbance, all of the baseline-corrected spectra were normalized as shown in Fig. 2(b). For normalization, the spectra were divided by the

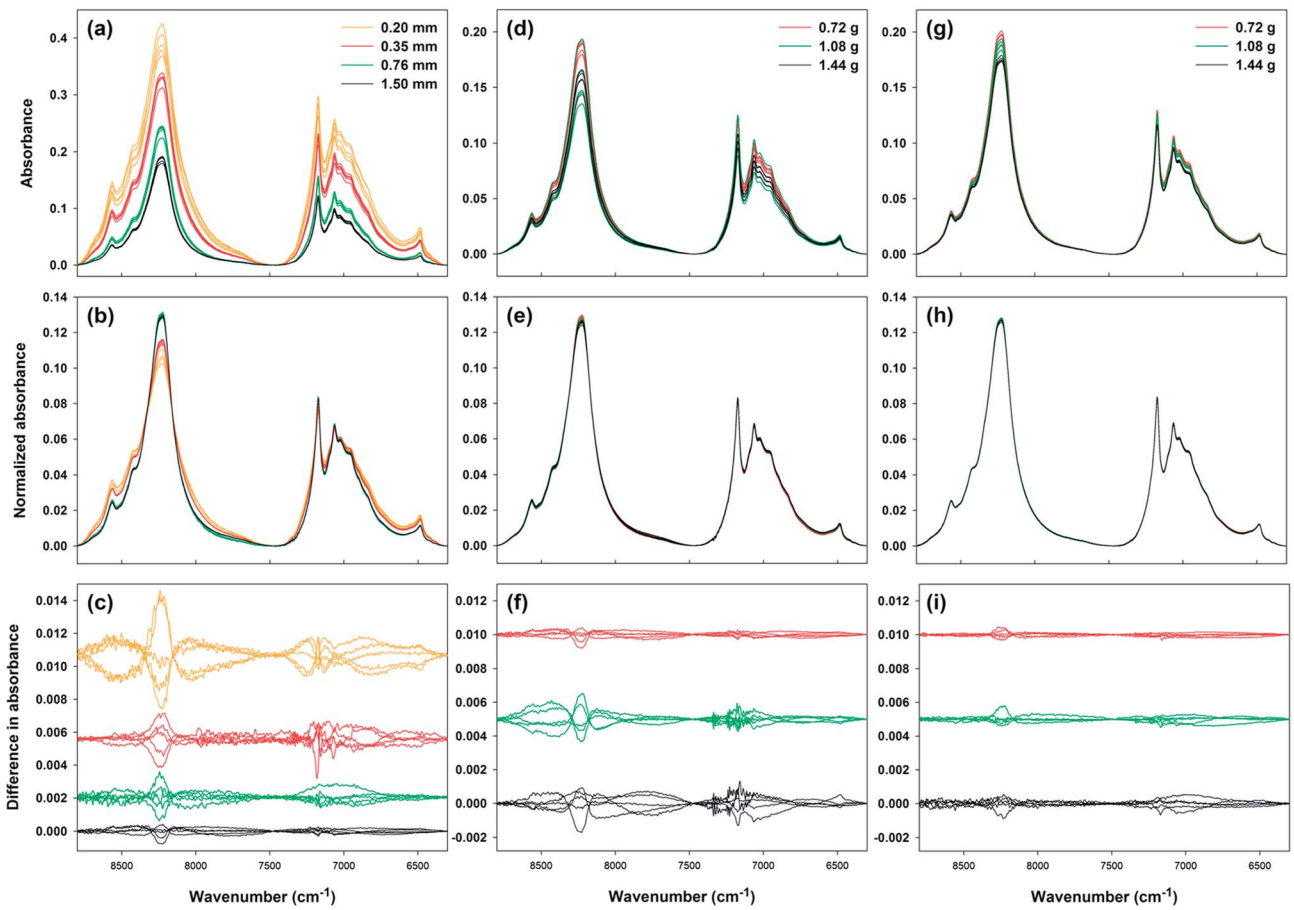

Fig. 2 Baseline-corrected 5 replicate NIR spectra (8800–6300 cm^{-1} range) of a PE pellet (density: 0.918 g cm^{-3}) measured in PTFE packings composed of four different sizes (0.20, 0.35, 0.76 and 1.40 mm, (a)), the packings with 0.72, 1.08 and 1.44 g of 1.40 mm-beads without (d) and with (g) the use of the diffuser. In each case, the corresponding normalized spectra ((b), (e) and (h)) and difference spectra ((c), (f) and (i)) are subsequently shown.

corresponding integrated area calculated in the 8800–6300 cm^{-1} range. The replicate spectral features in each case become more reproducible after normalization. They are most similar and dissimilar with each other in measurements based on 1.40 mm and 0.20 mm particle-packings, respectively. To effectively highlight the minute spectral variation, the difference spectra generated by subtracting each spectrum from the average spectrum of all the five spectra were examined as shown in Fig. 2(c). The difference spectra in each case were arbitrarily offset for comparison. The magnitude of the difference becomes smaller as the size of the particles in a packing becomes larger. For quantitative evaluation, the relative standard deviations (RDSs) of absorbance at 8230 cm^{-1} (the most intense peak) were calculated. The RSDs resulting from measurements with 0.20 (±0.04) mm, 0.35 (±0.05) mm, 0.76 (±0.05) mm, and 1.40 (±0.03) mm of particle-packing were 2.77%, 1.11%, 0.70%, and 0.34%, respectively. The use of smaller particles is obviously advantageous to increase the photon diffusion as well as the interaction with a pellet; however, the occurrence of excess diffusion simultaneously increases the attenuation of photons and complexity in photon migration in a packing, and is therefore disadvantageous in acquiring reproducible spectra.

After fixing the size to 1.40 mm, the amount of 1.40 mm beads in a packing increased from 0.72 g to 1.08 g, and 1.44 g (an increment of 0.36 g), and transmission spectra of the same PE pellet were collected as displayed in Fig. 2(d). Ninety beads are equivalent to 0.36 g. The *y*-span is approximately 50% smaller than that in Fig. 2(a). The magnitude of the 8230 cm^{-1} band varies when the amount of beads changes in a packing, while the spectral variation greatly decreases after normalization as shown in Fig. 2(e). As found in the corresponding difference spectra (Fig. 2(f)), the most reproducible measurement is achievable by the use of 0.72 g bead-packing. The RDSs of the absorbance at 8230 cm^{-1} for measurements with 1.08 g and 1.44 g bead-packings increased to 0.82% and 0.90%, respectively. The incorporation of redundant beads in a packing generates excess diffusion and negatively influences reproducibility.

There is a considerable void volume in the bead packing. Because of this, a small portion of the applied radiation propagates through the void space by successive reflections at or near the bead surface. This type of radiation, termed "non-fully diffused radiation" (NFDR), may not interact with samples. Therefore, NFDR could negatively influence the sample representation and measurement reproducibility. One simple approach to suppress NFDR is irradiation of the pre-diffused radiation to the bead packing rather than the collimated radiation. With this method, the probability of NFDR is greatly diminished.[24] For this purpose, the front glass window was replaced with a 0.8 mm-thick PTFE disk as a pre-diffuser, and the transmission spectra of the same PE pellet were collected again. Fig. 2(g) and (h) present the corresponding baseline-corrected and normalized spectra, respectively. As shown in the figure, the spectral variation among the replicate spectra in all three cases is further reduced by placing the diffuser. This result proves that the introduction of the pre-diffused radiation is effective in diminishing the chance of NFDR and eventually in further improving the reproducibility. Noticeable in the corresponding difference spectra (Fig. 2(i)), the RSD of the absorbance at 8230 cm^{-1} is minimized to 0.22% when the pellet is measured in the 0.72 g bead-packing with the diffuser in-place, while those from the measurements with the 1.08 g and 1.44 g bead-packing are 0.33% and 0.44%, respectively. Therefore, the diffuser-placed BITS (composed of 0.72 g 1.5 mm-PTFE beads) was solely employed for the subsequent measurements of the other PE pellets.

3.2 Spectral features of a PE pellet collected with BITS and the corresponding pressed PE disk

To examine the possible spectral distortion and bandwidth variation in the spectra collected using the diffuser-placed BITS, the standard transmission spectra were collected using a pressed PE disk and compared with those obtained using the scheme. Using the same low-density PE pellets, the flat disks with thicknesses from 0.5 mm to 4.0 mm (with an increment of 0.5 mm) were prepared by melting them above 140 °C and pressing the melt appropriately. Fig. 3(a) shows the baseline-corrected transmission spectra of the pressed PE disks with thicknesses varying from 0.5 to 4.0 mm. The absorbance increased with the thickness, and the variation was linear as shown in the inset, in which the absorbance at 8230 cm^{-1} is plotted *versus* the disk thickness. The average absorbance of the PE pellet at 8230 cm^{-1} obtained by the diffuser-placed BITS was 0.19 based on Fig. 2(g), so it was equivalent to the absorbance of a 1.8 mm-thick PE disk from the estimation based on the plot of the thickness *vs.* the absorbance. The estimated thickness corresponds to the average pathlength for all the interacted photons with the pellet.

Using the transmission spectrum of a 2.0 mm-thick PE disk, a new spectrum was generated by appropriate multiplication to match the spectral features acquired using the diffuser-placed BITS most closely, as shown in Fig. 3(b) for comparison. Black and red indicate the generated spectrum based on the PE disk and the pellet spectrum collected with the BITS, respectively. The features of the generated spectrum are mostly similar to those of the BITS-based spectrum; however, absorption of the 8230 cm^{-1} band is relatively smaller when the BITS is adopted. In the measurement of the PE disk, air was used as the background. When the NIR radiation passes the disk, the degree of scattering is slightly greater at higher wavenumbers (shorter wavelength), and so the amount of the corresponding photons reaching a detector decreases due to the additional attenuation by scattering compared to that in lower wavenumbers. This results in greater absorbance for bands in the higher wavenumber region. In contrast, since the same amount of beads, without including a pellet, was used as the background in the BITS, the different degree of scattering across the spectral range can be compensated for. Therefore, the absorbance of the 8230 cm^{-1} band is lower compared to that of the PE disk. Except for the difference in the absorbance of the 8230 cm^{-1} band, there is no significant difference in the spectral shapes such as the bandwidth between the two cases. A comparison revealed

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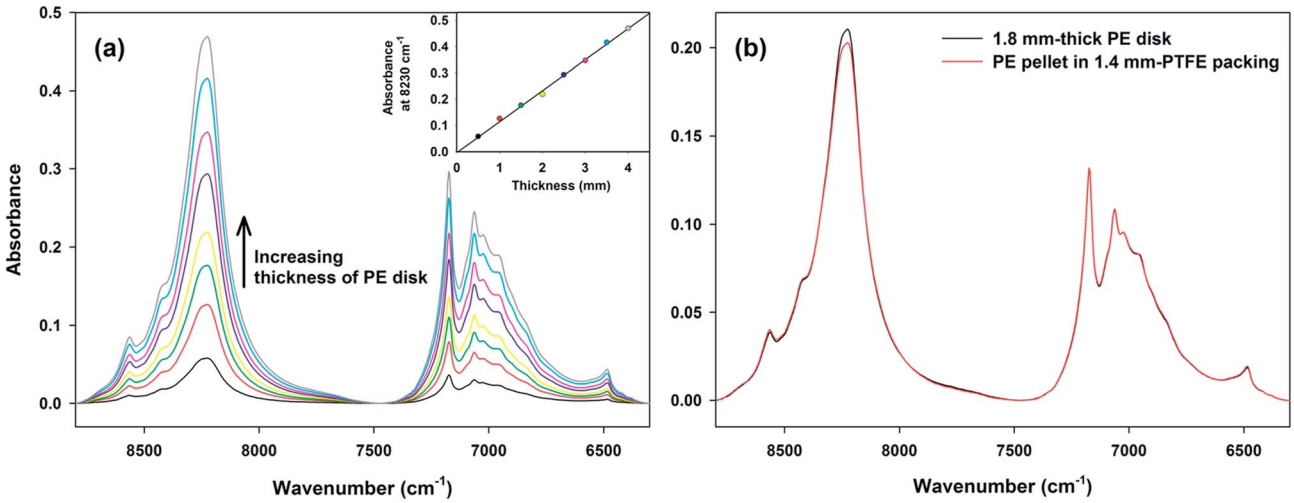

Fig. 3 Baseline-corrected NIR spectra of the pressed PE disks with thickness varying from 0.5 to 4.0 mm (a). The disks are prepared from low density pellets (density: 0.918 g cm^{-3}). The inset shows the correlation between absorbance at 8230 cm^{-1} and the thickness. Comparison of a generated spectrum corresponding to the 1.8 mm-thick PE disk (black) and the spectrum of the pellet acquired using the diffuser-placed BITS (red) (b).

that the diffuser-placed BITS produces reproducible transmission spectra of a PE pellet with correct representation of the polymeric structure.

3.3 Simulation of photon propagation through a packing of PTFE beads and particles

A numerical simulation based on the Monte-Carlo method was employed to investigate NIR propagation characteristics inside packings of PTFE beads and particles. As shown in our previous study,[24] individual photons were assumed to propagate elastically and independently through a packing in a random fashion in three-dimensional space. Incident radiation was assumed to have uniform intensity over the illuminated area (21 mm^2 (7.0 × 3.0 mm)) and equal probability of being injected at any point on the sample surface. Refractive indices used for PE pellets, PTFE and the surrounding area (such as void space) were 1.540, 1.350 and 1.000, respectively.

Fig. 4 shows the simulation results of top-view NIR photon (8230 cm^{-1}) propagations through the PTFE bead (1.40 mm) packing ((a)–(c)), the particle (0.35 mm) packing ((d)–(f)) and the diffuser-placed bead packing ((g)–(i)). Absorption coefficient of PE and PTFE at 8230 cm^{-1} for the simulation was set zero and 2 cm^{-1}, respectively. The absorption coefficient of PE was determined based on the correlation between the thickness of PE disk and absorbance as shown in Fig. 3(a). In each case, hundred thousand photons were introduced at three different locations: center, 3 mm left of center, and 3 mm right of center, as indicated by an arrow in each figure. The x-direction spanned 16 mm (from −8.0 to 8.0 mm), which was the actual window diameter of the sample cell. The circles correspond to the packed beads. The hexagon in each figure approximately mimics the shape of the PE pellets shown in Fig. 1. The lines indicate the paths of individual photon propagations, and the darker color at a given area implies a higher probability of photon crossing.

When the radiation was applied at the center of a bead packing (Fig. 4(a)), a large portion of photons migrated along the center line, and the photons were fairly homogeneously present inside the hexagon (pellet). In contrast, when photons were injected at 3 mm left of the center, the occurrence of photon migration was higher at the left part of the pellet (Fig. 4(b)) and *vice versa* (Fig. 4(c)). This indicates that the interaction of photons with a pellet is inhomogeneous in the case of off-center illumination. With smaller particle packing, the photons propagated over a larger area with a longer travel distance due to the higher degree of diffusion. Therefore, the chance of photon migration inside the hexagon increased, as indicated by the darker color. Inhomogeneous photon interaction with a pellet was also observed with off-center illumination (Fig. 3(e) and (f)).

In contrast, the placement of a diffuser (blue-colored layer) in front of the bead packing caused more diffused photon propagation and allowed photons to interact with more beads, as indicated by the increased number of darker circles (beads). The propagation of photons inside the hexagon was fairly homogeneous even in the case of off-center illumination. Illumination of a bead packing using pre-diffused radiation generated a more homogeneous photon presence around the pellet. More homogeneous photon interactions eventually generated more reproducible spectral features, as experimentally demonstrated in Fig. 2.

The reproducibilities of transmission measurements based on the three schemes shown in Fig. 4 were semi-quantitatively compared by evaluating the standard deviation of calculated absorbances at 8230 cm^{-1} obtained from 10 independent simulations. The averages and standard deviations of 10 calculated absorbances from propagation through the bead packing, the particle packing and the diffuse-placed bead packing were 0.2257 ± 0.0011 (0.49%), 0.2699 ± 0.0011 (0.41%), and 0.2687 ± 0.0007 (0.26%), respectively. The numbers in

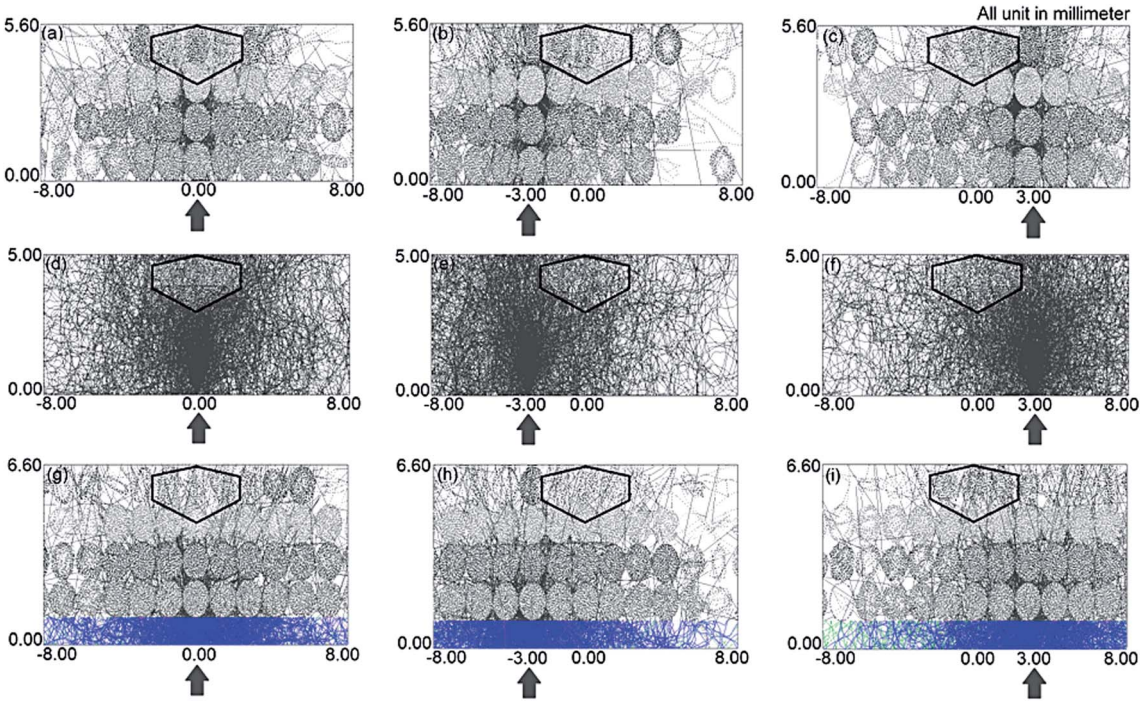

Fig. 4 Results of Monte-Carlo simulation showing top-view NIR photon (8230 cm^{-1}) propagations through the PTFE bead (1.40 mm) packing ((a)–(c)), the particle (0.35 mm) packing ((d)–(f)) and the diffuser-placed bead packing ((g)–(i)). The hexagon in each figure approximately mimics the pellet shape from Fig. 1. The arrows indicate photon illumination locations in each case.

parentheses indicate the corresponding RSDs. The standard deviation was smallest for measurements based on diffuser-placed BITS. Overall, the Monte-Carlo simulation results are in accordance with the spectral variation in experimentally acquired spectra, as shown in Fig. 2.

3.4 Determination of PE pellet density using diffuser-placed BITS

To build a PLS model for the determination of density, 10 pellets from each PE sample (a total of 20 calibration samples) were randomly selected. The NIR transmission spectra of individual pellets were collected (one spectrum per pellet) using the diffuser-placed BITS. Among 10 spectra of 10 pellets, only one spectrum was randomly chosen from each sample, giving a total of 20 spectra gathered as the calibration set. A PLS model was developed using the calibration set and the constructed model was used to predict the densities of 50 independent PE pellets (5 validation samples, 10 pellets per sample). The resulting prediction error is a gauge for the analytical utility of the proposed scheme, since it would be maintained at a low level if shape-tolerable reproducible spectra are collected. For further comparison, the number of pellets from each sample included in the calibration set was increased from 2 to 9, and the densities of the same validation pellets were predicted using corresponding PLS models. In all cases, 5 PLS factors were used for modeling.

Although only one pellet from each sample was included for the calibration, the standard error of prediction (SEP) in determining the densities of 50 validation pellets was 0.0008 g cm^{-3}, good and comparable to that achieved in our previous publication in which NIR transmission spectra were collected through packed PE pellets.[24] The result demonstrates that NIR spectral features acquired from one pellet are quite representative of other pellets for a given PE sample. When two pellets were included for the calibration, the SEP slightly decreased to 0.0006 g cm^{-3}. The inclusion of greater spectral variability from differences in the pellet shape allowed more accurate density determination for pellets with dissimilar shapes. When more than two pellets were included for the calibration, there was no significant improvement in SEPs. The overall result again confirms the analytical utility of diffuse-placed BITS for reproducible transmission spectral collection for samples with dissimilar shapes.

3.5 NIR measurement and mining locality identification of tourmalines with different shapes

NIR spectroscopic identification of tourmalines according to their mining locality was also attempted with the diffuser-placed BITS. The selection of tourmalines was intended to evaluate the analysis of samples with larger shape variability. As shown in Fig. 1, tourmaline appearances differ greatly, with no common shape. For the initial investigation of spectral features, one tourmaline from each locality was selected, and spectra were collected for each sample in triplicate. During replicate spectral collection, a tourmaline was re-loaded with random orientation in the bead packing.

Fig. 5(a)–(c) show raw transmission NIR spectra of tourmalines A, B and C, respectively, in the 8050–6500 cm^{-1} range. A

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picture of the measured tourmaline is shown in the corresponding figure. Spectral features of tourmaline A are clearly different from those of tourmalines B and C, indicating a significant compositional and structural dissimilarity. Only one small peak around 6940 cm^{-1} was observed on top of the background-like broad spectral features for tourmaline A. In contrast, the absorption bands of tourmalines B and C are distinct and fairly similar to each other. Therefore, differentiation between these two localities is more challenging and reproducible tourmaline spectra must be acquired. Tourmaline A was excluded from further study because it was easily discriminated by observing the spectral features.

To compare spectral features in detail, the baselines of spectra were linearly corrected at 8050, 7420 and 6500 cm^{-1}. The resulting baseline-corrected spectra were then normalized by dividing each spectrum by the corresponding integrated area calculated in the 8050–6500 cm^{-1} range. Fig. 5(d) and (e) show normalized spectra corresponding to the raw spectra in Fig. 5(b) and (c), respectively. After normalization, the spectral features of each sample were similar. The 7010 cm^{-1} band is from the first overtone of OH stretching, while the 7660 cm^{-1} band corresponds to a combination of the first overtone of OH stretching and Mg–OH bending.[27]

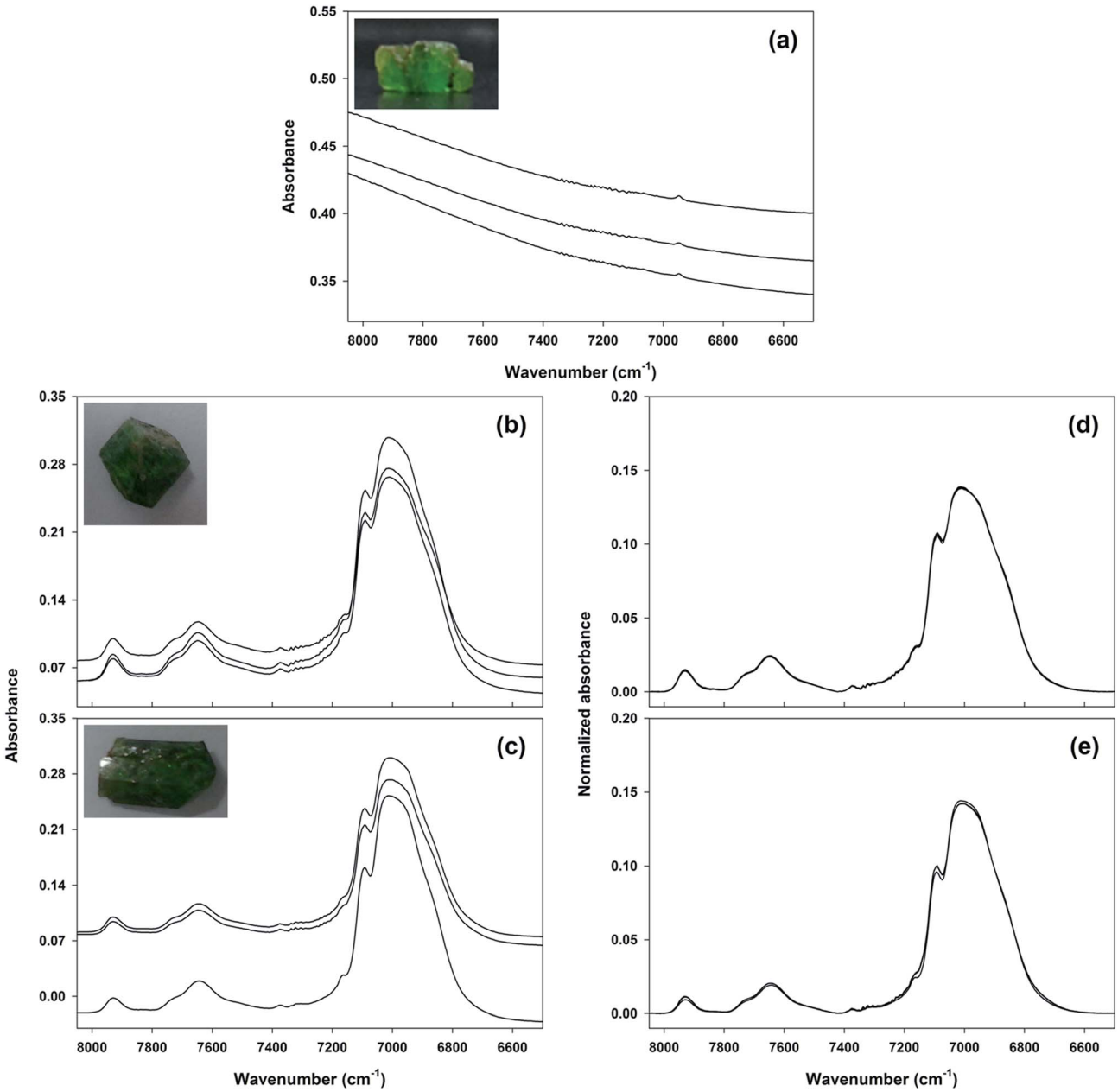

Fig. 5 Triplicate NIR spectra (8050–6500 cm^{-1} range) of tourmalines A (a), B (b), and C (c). A picture of the measured tourmaline is shown in each plot. Normalized spectra of tourmalines B (d) and C (e) are also shown.

Fig. 6(a) shows the average spectra of tourmalines B (red) and C (blue), which were produced by averaging 15 normalized spectra collected from 5 tourmalines of each locality. The shade with the same color indicates the corresponding standard deviation. The average spectrum of tourmaline C is arbitrarily offset for easier comparison. The difference spectrum between two average spectra (grey) is also present. Spectral features of tourmalines B and C are generally similar; while, there are small differences in the intensity and shape of absorption bands between tourmalines B and C, as revealed in the difference spectrum. In particular, the sharp spectral differences observed in the 7200–6900 cm^{-1} range are noteworthy. An explanation for the spectral differences requires extensive investigation based on geochemistry and analytical data. This investigation is beyond the scope of this study, which is mainly concerned with reproducible transmission spectral collection for samples of varying shapes. Nonetheless, it is clear that the spectral difference is due to compositional and structural dissimilarity caused by factors such as soil composition, temperature, pressure and the crystallization procedure.

To simultaneously compare the features of all studied tourmalines, PCA was performed using all of the spectra and the resulting scores were examined. Initial five factors described 99.4% of a total spectral variation present in the dataset. Fig. 6(b) shows a score scatter plot constructed using the second and fifth score. Red and blue scores correspond to tourmalines B and C, respectively. The numbers in the parenthesis in the *x*- and *y*-axis indicates the corresponding percentage of spectral description over the total variation. The boundary line between two groups was determined using linear discriminant analysis (LDA). The two groups are discernible as shown, although the two tourmaline B scores slightly cross the boundary to the other group. This publication used only 5 tourmalines from each mining locality as a feasibility study to evaluate the performance of diffuser-placed BITS. Future investigation of NIR spectroscopic differentiation according to mining locality requires

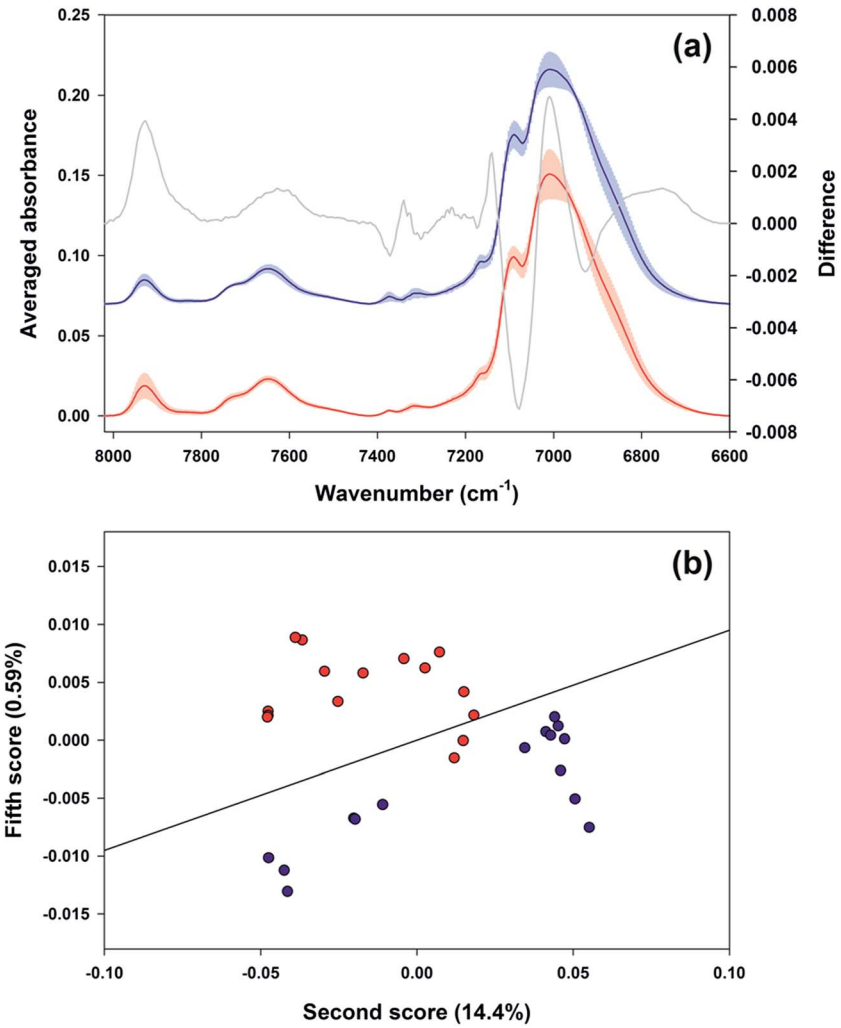

Fig. 6 Average normalized spectra for tourmalines B (red) and C (blue) (a). The shade with the same color indicates the corresponding standard deviation. The difference spectrum between two average spectra (grey) is also present. A score scatter plot constructed using the second and fifth scores with a boundary line determined using LDA is also shown (b).

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more representative samples with accurate compositional and structural information.

4. Conclusion

The diffuser-placed BITS, in which diffused radiation was applied around a sample using PTFE beads, was effective in collecting reproducible transmission NIR spectra of samples with inconsistent shapes. For a sample with varying shapes, the sample is usually cut perpendicularly and the cut faces are polished for reliable transmission measurements. This destructive sample preparation is not required with the proposed scheme, and spectral collection is simple. We think that the scheme has further potential to acquire reproducible transmission spectra for other samples. For the measurement of different samples, optimal experimental parameters such as bead size and packing thickness would be searched for again in conjunction with sample shape and size. The expandability of the scheme for the analysis of diverse granular samples is in progress.

Acknowledgements

This research was supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education, Science and Technology (2012R1A1B3003965).

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